THE TANGENT SPACE AT A POINT WITH SEPARABLE RESIDUE FIELD EXTENSION AS THE DUAL OF THE COTANGENT SPACE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement ends with the author's own remark that the result holds more generally for a separable algebraic residue field extension, and the proof is carried out in that generality; both are reproduced as written.

Editorial note. The counted spans carry 6 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-varieties/retained-preamble.tex lines 1-77), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-tangent-space-rational-point is defined only in varieties.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Duality between tangent space and fibre of the cotangent module. Same chapter, Lemma lemma-tangent-space-cotangent-space, invoked at proof lines 3080-3081. Supplies the second isomorphism from the first.
- The fibre of the relative differentials depends only on the fibre, and the stalk of the differentials on the local ring. Morphisms, Lemma morphisms-lemma-base-change-differentials and Modules, Lemma modules-lemma-stalk-module-differentials, invoked at proof lines 3082-3085. Supplies the reductions to the spectrum of the local ring of the fibre.
- Transitivity exact sequence for the cotangent complex and its behaviour under filtered colimits. Algebra, Lemma algebra-lemma-exact-sequence-NL and Lemma algebra-lemma-colimits-NL, invoked at proof lines 3095-3101. Supplies the reduction to the vanishing statements for the residue field over k.
- A finite separable field extension is etale. Algebra, Section algebra-section-etale, invoked at proof lines 3102-3104. Supplies the vanishing of the cotangent complex of the residue field over k.

Source reference. Original lemma 07BQ: sources/stacks-chow/tags.txt, source lines 9350–9350. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 00S2: sources/stacks-chow/tags.txt, source lines 1024–1024. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original section 00U0: sources/stacks-chow/tags.txt, source lines 1094–1094. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0B2D: sources/stacks-varieties/pinned-original-source.tex, source lines 3014–3014. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 08TE: sources/stacks-chow/tags.txt, source lines 11162–11162. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01V0: sources/stacks-chow/tags.txt, source lines 2429–2429. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes. Let $x \in X$ be a point and let $s = f(x) \in S$. Assume that $\kappa(x) = \kappa(s)$. Then there are canonical isomorphisms*

$$\mathfrak{m}_x/(\mathfrak{m}_x^2 + \mathfrak{m}_s\mathcal{O}_{X,x}) = \Omega_{X/S,x} \otimes_{\mathcal{O}_{X,x}} \kappa(x)$$

and

$$T_{X/S,x} = \mathrm{Hom}_{\kappa(x)}(\mathfrak{m}_x/(\mathfrak{m}_x^2 + \mathfrak{m}_s\mathcal{O}_{X,x}), \kappa(x))$$

This works more generally if $\kappa(x)/\kappa(s)$ is a separable algebraic extension.

Proof. The second isomorphism follows from the first by Lemma 0B2D. For the first, we can replace S by s and X by X_s, see Morphisms, Lemma 01V0. We may also replace X by the spectrum of $\mathcal{O}_{X,x}$, see Modules, Lemma 08TE. Thus we have to show the following algebra fact: let $(A, \mathfrak{m}, \kappa)$ be a local ring over a field k such that κ/k is separable algebraic. Then the canonical map

$$\mathfrak{m}/\mathfrak{m}^2 \longrightarrow \Omega_{A/k} \otimes \kappa$$

is an isomorphism. Observe that $\mathfrak{m}/\mathfrak{m}^2 = H_1(NL_{\kappa/A})$. By Algebra, Lemma 00S2 it suffices to show that $\Omega_{\kappa/k} = 0$ and $H_1(NL_{\kappa/k}) = 0$. Since κ is the union of its finite separable extensions in k it suffices to prove this when κ is a finite separable extension of k (Algebra, Lemma 07BQ). In this case the ring map $k \to \kappa$ is étale and hence $NL_{\kappa/k} = 0$ (more or less by definition, see Algebra, Section 00U0). □